

THE CALCULATION OF LARGE FAST REACTORS

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1. Introduction

The development of large fast power reactors is one of the major topics of the atomic energy programme of several countries. As far as the prediction of the physical behaviour of a fast breeder is concerned, the results presented at the Argonne Conference in 1965 showed no satisfactory agreement. To clear up the situation, the fast reactor teams should give a clear presentation of the methods and basic data they use for the theoretical design work. In this paper the fundamental procedure used in Karlsruhe in calculating fast reactors is presented. The problems which arise with the preparation of multi-group constants, the determination of reactivity coefficients with the calculation of second criticality and the long-time behaviour of fast reactors are discussed in some detail. A comment to each topic is given on the necessity of further improvements. It should be possible to calculate fast reactors as accurately as thermal reactors in the near future, say, for instance, to 1% accuracy in k_{eff} . As an important consequence the competition of safety characteristics and economy may be judged in a more reliable way.

2. Cross-sections and Group Constants

Microscopic Data

The basic nuclear data of the important materials for the calculation of fast and intermediate reactors are stored in the Karlsruhe nuclear data file KEDAK. The first KFK-120 data have been revised and updated.⁽¹⁾ A second edition is being prepared. The resonance data of the important fissile and fertile materials have been determined by analysing the known experiments⁽²⁻³⁾ and stored together with the important resonance data for structural and coolant materials in KEDAK.

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There are continuous improvements, updating, and complements of the nuclear data in Karlsruhe. Special interest should be given in general to the fission cross-section of Pu^{239} , the absorption cross-section of U^{238} , Fe, and Ni and the inelastic data for Pu^{239} . Also data of the higher Pu-isotopes are not up to date. Recommended data of the higher Pu-isotopes will be prepared for the KEDAK file by the group of S. Yiftah (Israel) in contract with the Fast Breeder Association Euratom—Karlsruhe. In Karlsruhe a reinvestigation of fission product cross-sections has recently been carried out⁽⁴⁾ for steam cooled reactors. The influence of the uncertainty of mentioned cross-sections on integral fast reactor quantities is given in refs. (5–6). These data may influence the safety and breeding features significantly.

Preparation of Group Constants

Taking the revised data of KFK-120 as a basis, the first Karlsruhe group constant set was prepared with a weighting spectrum typical for a 1000 MW(e) sodium cooled fast reactor. In order to use the Russian 26-group constants⁽⁷⁾ for the less important materials, the boundaries of the Russian set were adopted. The procedure of preparation and some characteristics of this set are given in refs. (5) and (8). The comparison of theoretical predictions for some integral data with experimental results of critical assemblies (ZPR/III, ZPR/VI) gives satisfactory agreement in critical mass and spectral indices⁽⁹⁾.

The preparation of group constants is now being undertaken in the main programme MIGROS including the part RESI in which the resonance self-shielding factors are determined.

Resonance Self-shielding

From the Russian group set we adopted the definition of self-shielding factors f_i , following the concept of factorizing the infinite dilution cross-sections and f_i -factors dependent on the temperature and material composition. The shielding factors are tabulated for certain temperatures and values of the cross-section σ_0 , dependent on the material composition. The shielding factors are determined before every reactor calculation by a suitable interpolation formula. Different definitions of the composition cross-section σ_0 have been investigated, but practically they result in negligible changes of f_i -values.⁽¹⁰⁾

The assumption of narrow resonance theory for the self-shielding effects of the heavy elements is quite satisfactory.⁽¹¹⁾ The resonance overlap effect of the same series gives sometimes remarkable contributions to the effective cross-sections. For resolved resonances the appropriate method is to produce the temperature-dependent cross-section from resonance data and using numerical integration to get f_i -factors. In this energy region the J-function method leads to difficulties. In the statistical region there is only a weak

dependence on the statistical parameters and this makes the use of f -factors very attractive because the infinite dilution values may be taken from the best known experimental results.

Elastic Moderation

In addition to the investigations concerning the calculation of elastic removal cross-sections described in ref. (5) two different weighting spectra for the elastic removal cross-sections have been used for core and blanket.⁽¹²⁾ The results for a spherical reactor with core composition according to case No. 8 of ref. (8) without fission products are summarized in Table 1. The

TABLE 1. INFLUENCE OF TWO DIFFERENT WEIGHTING SPECTRA IN CORE AND BLANKET

	Core weighting spectrum only	Different core and blanket spectra
k_{eff}	$0.9996 \pm 1 \times 10^{-4}$	$0.9994 \pm 1 \times 10^{-4}$
$\Delta k_{\text{Na void}} (50\% \text{ voided})$	$0.0158 \pm 2 \times 10^{-4}$	$0.0156 \pm 2 \times 10^{-4}$
BR_{int}	1.073	1.071
BR_{ext}	0.394	0.395
BR_{tot}	1.467	1.466

values for the internal and external breeding ratios and the reactivity change for 50% loss of sodium out of the core remain practically unchanged if an extra weighting spectra is used for the blanket. But the power distribution in the blanket has increased by about 4%. Compared with the procedure given in ref. (5) a better description of the anisotropic scattering in the centre of mass system is now being investigated. Most of the angular distributions are taken from ref. (13) together with more recent data.

In conclusion it can be said that the effect of different weighting spectra (i.e. different group sets) in the zones of a multi-layer fast reactor is not negligible for a proper description of the power distribution and of the blanket management and blanket optimization features. Also the radiation exposure to the adjacent steel vessel is influenced by using different sets. This effect is confirmed by other methods, which will be discussed in section 2. The results require special attention to the space dependency of weighting spectra. Especially the use of fundamental mode calculations with many energy groups as in ELMOE-type codes has to be reinvestigated according to space dependent problems. For routine calculations it is felt that the description of elastic slowing down as given in ref. (5) is sufficient. For the reactor chosen to be built more refined methods describing elastic moderation

have to be used. In contrast with the Fast Breeder Association Euratom—Karlsruhe, the group of Professor Yiftah in Israel is investigating a semi-analytical representation of the collision density in the Doppler energy range for pure elastic down-scattering. The more general space dependent problem is being studied in Karlsruhe.

Transport Cross-section

The problems related to the appropriate weighting and shielding of the transport cross-section are well known.⁽¹⁴⁾ Here another effect is mentioned which may be significant in some cores.⁽¹⁵⁾ If one considers the space independent slowing-down mechanism in a consistent P_1 -approximation, the evaluation of the linear anisotropic scattering-down integral

$$\int_E^{E/\alpha} K_1(E', E) j(E') dE',$$

results for $\alpha < 1$ (say $A > 10$) in different transport cross-sections for different assumptions on the energy dependence of $j(E)$. Here $\alpha = (A-1)^2/(A+1)^2$, $j(E')$ = net current, $K_1(E', E)$ = linear anisotropic moment of the scattering kernel for neutrons scattered by atoms at rest. Assuming, for simplicity, isotropic scattering in the centre of a mass system and a smooth variation of the scattering cross-section $\Sigma_s(E)$ we have for one isotope A

$$\Sigma_{tr}^0 = \Sigma_a + \Sigma_s \left(1 + \frac{2}{3A \cdot \alpha} \right) \quad \text{for } \frac{j(E)}{E} \quad \text{as slowly varying function,}$$

$$\Sigma_{tr}^1 = \Sigma_t \quad \text{for } j(E) \quad \text{as slowly varying function,}$$

$$\Sigma_{tr}^2 = \Sigma_a + \Sigma_s \left(1 - \frac{2}{3A} \right) \equiv \Sigma_a + \Sigma_s (1 - \bar{\mu}) \quad \text{for } j(E) \cdot E \quad \text{as slowly varying function.}$$

In a fast sodium cooled reactor the current is roughly proportional to E up to 1 keV, then roughly constant up to some 100 keV. Because the assumptions made above are true in the range below 50 keV for C, O, Na, Fe, Ni, we see that the leakage due to the light materials is reduced in the lower energy groups, where, for instance, the Doppler effect is important. The mentioned light materials contribute more than 50% to the total cross-section in this energy range. It has been noted that the transport cross-section is very sensitive to the assumed energy dependence of the current. The effect is important especially for a steam cooled oxide breeder. In this case the transport cross-section increases roughly proportional to $1/\sqrt{E}$ when approaching lower energies. Again as in the preceding section and even more severely the space dependency of weighting spectra plays a role.

3. Methodical Procedure for Neutronic Calculations

Codes in Operation in Karlsruhe

Most of the codes used in Karlsruhe are combined in the master programme NUSYS.⁽¹⁶⁾ The subcodes are compatible in their formats so that they can always be arranged for the special purpose wanted. NUSYS contains more or less all necessary codes requested for the calculation of fast reactors (1-, 1½-, and 2-dimensional diffusion, 1-dimensional S_N , perturbation, Doppler effect routine, reaction rates, breeding ratios, heterogeneity corrections for the calculation of critical assemblies), burnup codes for a point model and 1½ dimensions are in operation.

Transport codes are needed in the calculation of large fast reactors only for the determination of blanket properties (about 7% change in the power distribution for the middle of the blanket is observed going from diffusion to S_4 approximation) or for a final prediction of design values. In this context also the evaluation of Monte Carlo codes⁽¹⁷⁾ is interesting.

Some Numerical Questions

In view of the discrepant results presented by the intercomparison in Argonne⁽⁸⁾ it was felt necessary to prove the insensitiveness of typical integral fast reactor quantities to the number of meshpoints and accuracy of source distribution used in a reactor calculation. For this investigation two different one-dimensional diffusion codes were used.⁽¹⁸⁻¹⁹⁾ In the MGP-code the diffusion equation is solved by quadrature, in the Diff (NUSYS)-code difference techniques are used. For a sodium cooled fast reactor (No. 9 of ref. (8)) a reduction from 95 core points and 45 blanket points to 30 and 15, respectively gives errors in breeding ratio and reactivity change for total Na loss out of the core which are less than 0.2%. (In all cases k_{eff} was accurate to 10^{-4} .) This error is increased for the softer spectrum so that in the region of large cross-sections the flux may become strongly negative for a small number of meshpoints. Improving the accuracy of the source distribution from 10^{-2} to 10^{-5} for two consecutive iterations practically no change in local source and fission density and breeding ratio was observed. For routine calculations of fast sodium cooled reactors only a small number of meshpoints is required.

Comparison of One- and Two-dimensional Methods in Calculating a Cylindrical Reactor

Reactor No. 9 of ref. (8) was investigated. For all calculations the Karlsruhe group set KFK 26-10 was condensed to four groups always weighting with the same spectrum. k_{eff} and $\Delta k_{\text{Na-loss}}$ (total sodium loss of core) have been

calculated with the two-dimensional diffusion code DIXY (NUSYS)⁽²⁰⁾ (about 1500 meshpoints have been used).

Using the one-dimensional diffusion code Diff (NUSYS), the reflector savings and generated group-dependent bucklings were determined successively. By interpolation of the corresponding k_{eff} -values the approximated k_{eff} -values were obtained.

	1-D. Diff	2-D. Diff
k_{eff}	$1.0073 \pm 1 \cdot 10^{-4}$	$1.0065 \pm 1 \cdot 10^{-4}$
$\Delta k_{\text{Na-void}}$	$0.0430 \pm 2 \cdot 10^{-4}$	$0.0428 \pm 2 \cdot 10^{-4}$

There is a good agreement between the results of the one- and two-dimensional methods. Also the breeding ratios show only small differences: about 1% in BR_{int} and 3% in BR_{ext} . The deviations of flux shape factors resulting from one- and two-dimensional methods do not exceed 5%.⁽²¹⁾

A more convenient method for simulating cylindrical reactors is the use of flux synthesis methods, either using the Ritz- or the Kantorowitsch procedure. An automated method using a code for the simplest case (separation of variables) is almost complete in Karlsruhe.⁽²²⁾

Investigation of Different Condensation Spectra and Different Numbers of Condensed Groups

In calculating two-dimensional problems it is necessary in almost all cases to collapse drastically the original group set because of the limited capacity of the computer. Many problems arise in finding the most suitable condensation spectra. In order to get an estimate of the error involved in this procedure, two different fundamental mode spectra were used (spectrum 1: $B^2 = 0$; spectrum 2: $B^2 = 7.38 \times 10^{-4}$) collapsing the 26-group set to 13, 10, and 5 groups. The resulting group cross-sections were used to calculate a spherical reactor, No. 9 of ref. (8), without changing the geometry.

k_{eff} and the breeding ratios (BR) are not very sensitive to group collapsing. The deviations from the case of 26 groups are small (spectrum 1, maximum deviations: $\Delta k = 0.5\%$, $\Delta \text{BR}_{\text{int}} = 1\%$, and $\Delta \text{BR}_{\text{ext}} = 3\%$). The deviations of values corresponding to spectrum 2 are generally smaller (maximum deviations: $\Delta k = 0.3\%$, $\Delta \text{BR}_{\text{int}} = 0.2\%$, $\Delta \text{BR}_{\text{ext}} = 2.5\%$).

A greater sensitivity is shown for the zero of the local sodium void coefficient (variations up to 3 cm) and the power distributions mainly in the blanket. The deviations in the power distribution are positive or negative depending not only on the spectrum used for condensation but also on the manner of combining the groups.

This means that for a rough estimate of integral values it is possible to use condensed cross-sections, however for spatial dependent values (danger coefficients) one should be more careful. These effects are more pronounced in multi-zoned reactors, especially in the investigation of blanket properties

(i.e. external breeding ratios, breeding and power distribution). To get reasonable accuracy in the blanket power distribution it is necessary to use two blanket condensation spectra.⁽²⁵⁾

4. Reactivity Coefficients

Coolant Loss

The reactivity change due to coolant loss is very sensitive to the basic data and methods used. In Karlsruhe for routine calculations the same group-constant set may be used for the two necessary criticality calculations (but with changed shielding factors according to the different material compositions in the normal case and in the loss case) or two different group sets (different in the used weighting spectrum for the two cases) may be used as described in ref. (5).

For very specific determination of the coolant loss coefficients one has to describe the slowing-down process properly by using ELMOE-type or even improved ELMOE-calculations. But the intercomparison⁽⁸⁾ shows that even results obtained with these refined methods may be discrepant by about 300% due to the different basic data involved. Furthermore, attention has to be paid to the fuel temperature of the voided region. This is especially significant for steam cooled reactors.⁽²³⁾ In testing the calculation procedure against experimental results of critical assemblies one has to be very careful; especially heterogeneity effects have to be considered properly. The sign of the reactivity coefficient may change in different arrangements of the fuel and coolant plates.⁽²⁴⁾

Doppler Coefficient

In Karlsruhe a code for the calculation of the Doppler effect (DCP) has been established on the basis of perturbation theory. The Doppler coefficient (DC) is given by

$$DC = - \sum_i \int (\phi_i^+ \frac{d}{dT} [\tilde{\Sigma}_{gi} + \tilde{\Sigma}_{fi}] \phi_i - \phi_i^+ \chi_i \sum_k v_k \tilde{\Sigma}_{fk} \phi_k) dV. \quad (1)$$

$\tilde{\Sigma}_{xi}$ is the effective group cross-section of type x in group i . The method of calculating $d\tilde{\Sigma}/dT$ is derived in such a way that it calculates in a short time the correct $d\tilde{\Sigma}/dT$ for sodium cooled fast breeder reactors now under discussion.

The method of calculation of

$$\frac{d}{dT} \tilde{\Sigma}_x = \frac{d}{dT} \left\{ \left(\int \frac{\Sigma_x}{\Sigma_t} dE \right) : \left(\int \frac{1}{\Sigma_t} dE \right) \right\} \quad (2)$$

is based in principle on R. B. Nicholson's paper,⁽²⁶⁾ where only statistical

parameters are used. For the determination of the integrals $\int \Sigma_x / \Sigma_t dE$ and $\int 1 / \Sigma_t dE$, we have split the denominator Σ_t into two parts. The first part contains the main contributions to Σ . These are the large resonances of the first U^{238} series ($\Sigma_{res}^{U^{238}}$) and an energy independent part, representing a kind of averaged cross-section of all other materials (C_x). The second part of Σ_t describes the deviation $\Sigma^{rest} = \Sigma_t - \Sigma_{res}^{U^{238}} - C_x$.

In the important energy regions of the fast reactors considered, the second part of Σ_t is small nearly everywhere compared with the first part so that the following expansion is allowed:

$$\frac{1}{\Sigma_t} = \frac{1}{\Sigma_{res}^{U^{238}} + C_x + \Sigma^{rest}} \approx \frac{1}{\Sigma_{res}^{U^{238}} + C_x} \left(1 - \frac{\Sigma^{rest}}{\Sigma_{res}^{U^{238}} + C_x} + \dots - \dots \right) \quad (3)$$

This expansion is inserted into the integrals

$$I_1 = \int \frac{\Sigma_x^{U^{238}}}{\Sigma_t} dE, \quad I_2 = \int \frac{\Sigma_x^{REST}}{\Sigma_t} dE \quad \text{and} \quad I_3 = \int \frac{1}{\Sigma_t} dE \quad (4)$$

($\Sigma_x^{U^{238}}$ indicates the contribution of the first U^{238} series to Σ_x and Σ_x^{REST} indicates all other contributions to Σ_x). Equations are obtained for the determination of C_x by the condition that the integrals of the first expansion term of I_k exactly disappear.

For I_1 and I_3 the C_x are equal and have the form

$$C^{I_{1,3}} = \frac{1}{\Delta E} \int (\Sigma_t - \Sigma_{res}^{U^{238}}) dE, \quad (5)$$

while for I_2

$$C^{I_2} = C^{I_{1,3}} + \frac{\int \Sigma_x^{REST} \Sigma_{res}^{REST} dE}{\int \Sigma_x^{REST} dE} - \frac{1}{\Delta E} \int \Sigma_{res}^{REST} dE. \quad (6)$$

Regarding the repulsion effect of resonances within the same series, we can carry out a second evaluation so that we have one resonance in the denominator of the expressions $1/(\Sigma_{res}^{U^{238}} + C_x)$ only. The overlap effect of resonances of the same series is so taken into account in the first-order term of this expansion. With these two evaluations the upper integrals and their temperature derivatives can be expressed by J -functions averaged over all possible neutron widths and terms of the form $\int \Sigma_x \Sigma_{res} dE$ and their corresponding temperature derivatives. The J -function $J = \int \psi / (\beta + \psi) dx$ is available in the form of a table. This procedure results in a relatively short computing time for the calculation of Doppler coefficients of fast reactors.

As far as the validity of the assumptions is concerned, an adequate number of resonances must be in a group, because only statistical parameters are used. This is, for example, not the case in the Russian group constants⁽⁷⁾ for energy groups lower than 100 eV. But this region is not very important for the Doppler coefficient of fast reactors, at least for the sodium cooled

type. Besides this another limitation exists, caused by the first type of evaluation. This limitation expresses that no Doppler material besides U^{238} must have such a nuclear concentration that the resonances of this material are higher than the average total cross-section. These two conditions are fulfilled in the presently discussed fast breeder reactors. Comparisons of the results with other teams who calculated Doppler coefficients showed no remarkable differences. These comparisons were done for the $d\Sigma/dT$ and for the total Doppler coefficients in ref. (8) with the Israeli team.

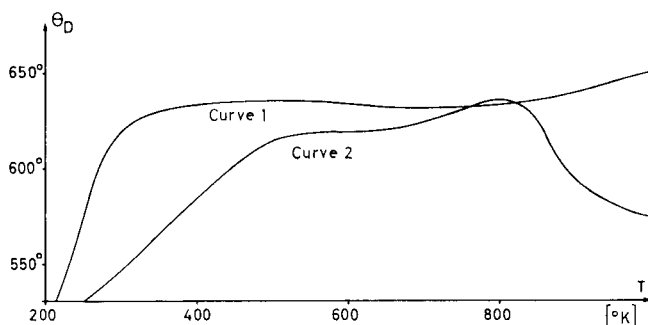


FIG. 1. Debye temperature θ as a function of temperature.

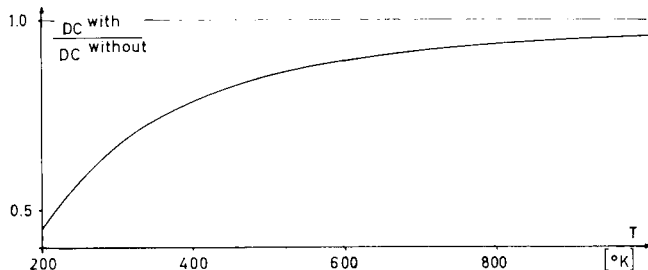


FIG. 2. Solid state correction factor for the DC.

The overlap effect of resonances of different series, which is included in the calculation method, shows no remarkable influence on the Doppler coefficient.⁽²⁷⁾

Besides the DCP programme there exists a first version of a programme for the interpretation of reaction rate measurements.

The influence of solid state properties on the DC^(28,29) can be taken into account by a correction factor.⁽²⁸⁾ If more accurate measurements of the crystal-frequency spectrum of Dolling *et al.*⁽³⁰⁾ are used, the result is that a fictitious Debye temperature of about 600°K can describe the solid state influence on DC in the interesting temperature range.⁽³¹⁾ Curve 1 of Fig. 1

shows the Debye temperature which gives the temperature dependent correction factor between DC with solid state properties and DC without solid state properties. The curve 2 shows the Debye temperature which describes the specific heat in a correct way. One recognizes a strong dependency of the Debye temperature on the temperature and how it is calculated. Figure 2 shows the magnitude of this correction factor ($DC^{with}/DC^{without}$).

In conclusion it may be said that the calculation of the Doppler coefficient gives satisfactory results. Differences mainly result from the use of different basic group sets. Assuming $1/T$ dependency of the DC, there is also good agreement between successive k calculations and the method described above. Therefore, it seems that a special DC code for the resolved resonance region is not necessary, as codes (RESI) are available for the temperature dependent group constants in this range.

5. Investigation of Core Disassembly and Second Criticality

General Remarks

The safety analysis of large reactors is usually divided into two parts:

1. As a model for the accident situation a time-dependent reactivity insertion, e.g. reactivity ramp, is evaluated.
2. The energy release is calculated according to the reactivity input.

The first problem may be solved by using the FORE code or analogue computer techniques. The discussion will be restricted to the second point.

Since the analysis of possible accident situations is widely based on unknown and hypothetical features, it is reasonable to study the effect of the following parameters:

1. Doppler coefficient (DC);
2. rate of reactivity ramp;
3. primary flux;
4. delayed neutrons.

Furthermore, there is much interest in the study of the effects of zoned and non-spherical reactors. The investigation of the latter points will be carried out later.

Bethe-Tait Model Containing Doppler Feedback

The first four points may be studied in a modified Bethe-Tait formalism in connexion with a spherical symmetrical core. Although some investigations⁽³²⁻³⁵⁾ have been carried out in this field, it is necessary to elaborate the model as we will see below.

The starting point of the analysis⁽³⁶⁾ is a modified Bethe–Tait method. The separation into a time dependent and a space dependent function is established and perturbation theory is applied. Thus, the problem is reduced to a system of six non-linear, ordinary differential equations of the first order for the following variables:

1. energy;
2. flux;
3. delayed neutron precursor;
4. Doppler feedback;
5. feedback by expansion of the core;
6. first derivative of the feedback by expansion.

The integrals over the space variables, which occur in the differential equations, can be solved analytically because the energy dependence of the pressure and the radial dependence of flux and importance are approximated by polynomials. Results for vanishing DC are very easily understood. The destructive energy increases, if the rate of the reactivity ramp is augmented or the primary flux decreases. For negative DC the flux is not monotonic during the excursion, since the Doppler feedback induces some oscillations. Now, the released energy depends very strongly upon the phase of such a cycle, at which the excursion is ended by expansion. Hence, a change of, for example, 5% in the DC may involve a factor of 6.

The Relative Maximum of Released Energy

The values for the released energy of refs. (33–34) do not show any monotonic behaviour, and for this reason the DC was switched off temporarily in ref. (35). But the latter approximation induces a large overestimate for the released energy. Furthermore, the influence of delayed neutrons should be estimated.

Therefore, the conditions for the occurrence of the relative maximum in the released energy have been investigated.⁽³⁶⁾ A code named FAUN has been elaborated which incorporates this result. FAUN evaluates upper limits for the energy and the latter are also valid if the reactor parameters vary by some amount because of the uncertainty in the starting conditions.

With this code the influence of the parameters was checked numerically for a meltdown accident of a sodium cooled fast reactor (Na 1) for several ramp rates. The results in Table 2 show the features predicted by theory for a ramp rate of \$100/sec. In this table we listed the thermal energy above the boiling point, which is an upper limit for the destructive energy. The energy decreases with increasing DC. The dependence on the reactivity ramp and primary flux is very similar to the case of vanishing DC.

In this context the effect of delayed neutrons is very interesting. In any

case, they reduce the energy but there are two different reasons for vanishing DC on the one hand and for large negative coefficients on the other. In the first case the reduction arises from the time difference between delayed critical and prompt critical and is of the order of some 10% (region above the intermediate line in Table 2). In the latter case the delayed neutrons damp the oscillations mentioned above and this effect leads to an essential reduction in energy (region below the intermediate line in Table 2).

TABLE 2. DESTRUCTIVE ENERGY IN KG TNT FOR EXCURSIONS WITH A REACTIVITY INPUT \$100/SEC. EFFECTS OF DELAYED NEUTRONS INCLUDED

Doppler coefficient [10 ⁻⁶ /°K]	Φ_0 [cm ⁻² sec ⁻¹]			
	10 ⁷	10 ¹⁰	10 ¹³	10 ¹⁶
0	4110	3500	2750	1680
-0.5	1660	1260	815	292
-1	812	587	350	100
-2	263	182	103	11.4
-3	117	6.26	6.87	5.19
-4	3.31	3.54	2.29	2.01
-8	0.77	0.78	0.80	0.88

Equation of State

The equation for the saturation vapour pressure is valid in the temperature region below the critical point. From a temperature lying somewhat below the critical temperature onwards the pressure increases more rapidly due to the incompressibility of fluids.⁽³⁵⁾ In the code FAUN only the vapour pressure law has been used up to now. Therefore, only upper limits for the energy release are obtained in some cases. FAUN also takes into account that a certain amount of fuel is vaporized so that the Doppler effect is reduced by this effect.⁽³⁷⁾

6. Long-time Behaviour of Fast Power Reactors

General Remarks

The investigations of the long-time behaviour of fast power reactors includes the overall time behaviour caused by the fuel burnup and the fuel management. In this connexion two sets of problems are distinguished:

1. The time behaviour during one reloading cycle or only a few cycles, which is usually the object of fuel burnup studies, taking into consideration Δk_{eff} , power distribution, utilization of the fuel, etc.

2. The time behaviour of the isotopic composition of the fuel during a very great number of reloading cycles caused by recycling the Pu (externally closed fuel cycle).

Cyclic Loading Scheme of the Fuel Elements

The time behaviour for a burnup cycle, i.e. between two successive changes of fuel elements, is studied on the basis of a cyclic scheme for single groups of fuel elements in the core.⁽³⁸⁾ These groups are defined as follows:

1. n fuel elements of the same radial region r , which are not necessarily adjacent, are combined to a group.
2. In every reloading operation only one element of a group is changed.
3. The difference between the burnup of the single elements of a group is $1/n$ of the maximal flux time (these states should be realized or approximated by appropriate fuel management before the steady operation of the reactor).
4. The burnup of the n elements of a group after discharging one element and restoring the fresh state of a group (i.e. the averaged burnup of the group is a minimum) is approximately

$$A_1 = 0, \quad A_2 = \frac{1}{n} A_{\max}, \dots, \quad A_n = \frac{n-1}{n} A_{\max}$$

($A_{\max}$ = maximum fuel burnup in MWd/t). Then the averaged burnup of a group is given by

$$\bar{A}_{\min} = A_{\max} \frac{n-1}{2n}.$$

5. The burnup of the elements of a group before changing an element (the group is burnt out, i.e. the averaged burnup is a maximum) is approximately

$$A_1 = \frac{1}{n} A_{\max}, \quad A_2 = \frac{2}{n} A_{\max}, \dots, \quad A_n = A_{\max}.$$

Then the averaged burnup of a group is

$$\bar{A}_{\max} = A_{\max} \frac{n+1}{2n}.$$

6. Always in a reloading operation the element must be exchanged which has just reached the maximum burnup $A_{\max}$ or which would exceed $A_{\max}$ during the next burnup cycle.

The burnup cycle of the groups is shown in Fig. 3 schematically for $n = 3$. (The circles symbolize the fuel elements of a group, the numbers denote the burnup of the elements immediately after and before changing an element respectively.)

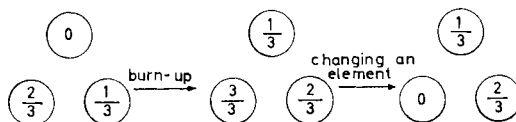


FIG. 3. Burnup cycle for a group of three fuel elements.

The real burnup state of the core varies between the two states $\bar{A}_{\min}$ and $\bar{A}_{\max}$. For calculating Δk_{eff} between the two extreme states, i.e. the maximum movement of k_{eff} caused by the burnup of the fuel, only the difference

$$\Delta \bar{A} = \frac{1}{n} A_{\max}$$

is important. Thus, by establishing groups of fuel elements in the specified manner the influence of the burnup is reduced to $1/n$ of the maximum burnup of the single fuel elements.

TABLE 3. THE DEPENDENCE OF Δk_{eff} AND THE UTILIZATION OF THE FUEL ON VARIOUS RELOADING ZONES

n	Δk_{eff} (%)		Utilization of the fuel (%)	Reloading cycle Δt [d]
	Without fission products	With fission products		
Zone I } $n = 3$ Zone II }	0.76	—	79	196
Zone I: $n = 3$ Zone II: $n = 4$	0.74	2.5	85	

The maximum Δk_{eff} and the reloading cycle Δt (corresponding to $A_{\max}$ and the power of the reactor) are determined by the choice of n . Δt is the reloading cycle for the central group at the position of the maximum neutron flux and hence for all groups in the whole core. Δt is given by T/n , T = life-time of the fuel element in the centre of the core.

The utilization of the fuel can be optimized by establishing several radial reloading-zones with various values for n . The influence of these space-dependent n on Δk_{eff} and the utilization of the fuel is shown in Table 3 for two reloading zones.

The results given in this table are calculated for the sodium cooled fast reactor study Na 1 with the Russian group set. The establishment of a second zone with $n = 4$ has no influence on Δk_{eff} , but improves the utilization of the fuel from 79% up to 85%.

The previous investigations of the time behaviour for one or a few burnup cycles are carried on and completed using a one-dimensional burnup code⁽³⁹⁾ taking into consideration the internal fuel management of the core and the blanket. In this code the z -dependent burnup is treated by an appropriate approximation. The purpose of further investigations is to find out appropriate design burnup and operation data which optimize the behaviour of fast reactors.

Long-time Behaviour of the Fuel Composition—The Steady State of Pu-Composition

The long-time variations of the isotopic composition of the fuel in fast reactors with externally closed fuel cycles are caused by the burnup of the

TABLE 4. STATIONARY PU-COMPOSITION Pu^∞ (at. %)

Reactor	Na		H ₂ O-steam	
Case	1	2	1	2
Pu 239	76.8	60.8	74.0	63.7
Pu 240	20.3	32.2	22.7	30.5
Pu 241	2.4	4.8	2.3	3.4
Pu 242	0.5	2.2	1.0	2.4

fuel and the external fuel management. In the course of time the isotopic composition of the fuel becomes stationary as is shown in ref. (40). The relative isotopic composition of the plutonium is the so-called Pu^∞ .

For the external fuel management the following cases are considered:

Case 1. Common reprocessing of the core and blanket elements and drawing off the breeding gain after reprocessing (case C in ref. (40)).

Case 2. Separate reprocessing of the core and blanket elements and drawing off the breeding gain out of the blanket (case B in ref. (40)).

For both cases there are different values for the Pu^∞ . Table 4 shows the Pu^∞ for a sodium cooled fast reactor and a steam cooled reactor of 1000 MW(e).⁽⁴¹⁾

It was found that Pu^∞ values are not very sensitive to the burnup and the loading scheme.⁽⁴²⁾

The study of the long-time behaviour of the fuel composition and all the problems connected with it is continued by considering the details of the

external fuel management and the blanket management, which may be important for the variation of the fuel composition.⁽⁴³⁾ In order to restrict the numerical efforts to a reasonable scale, which is necessary to study these problems, time averaged spectra for core and blanket regions are taken as a basis. The criticality of the reactor is approximately ensured by appropriate conditions for determining the new fuel composition after reprocessing. Also the breeding gain (positive or negative) is determined by these conditions.

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